

## What is `asyncio` in Python?

`asyncio` is a Python library used to write **concurrent code using the `async/await` syntax**. It enables **asynchronous programming**, which lets your program handle many tasks seemingly at the same time without needing multiple threads or processes.

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### Key points:

- **Asynchronous I/O:** Instead of blocking while waiting for operations like file reads, network requests, or timers, `asyncio` allows your program to **pause and switch to other tasks**, improving efficiency.
  - **Event loop:** At the core of `asyncio` is an event loop that manages and schedules asynchronous tasks.
  - **Coroutines:** Functions defined with `async def` that can be paused and resumed with `await`.
  - **Non-blocking:** Tasks run cooperatively; they yield control during `await` points, allowing other tasks to run.
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### Why use `asyncio`?

- Handle many **IO-bound tasks** efficiently without needing multiple threads.
- Write readable asynchronous code using `async / await`.
- Useful for network servers, web scraping, real-time data processing, or any task with lots of waiting.

## Simple example:

python

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```
import asyncio

async def say_hello():
    print("Hello")
    await asyncio.sleep(1) # non-blocking wait
    print("World")

asyncio.run(say_hello())
```

## Output:

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```
Hello
# (waits 1 second without blocking)
World
```